

Chirag Singh

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Full-Stack Software Engineer with **3.5+ years of experience** designing and delivering scalable web and cloud applications using **.NET (C#)**, **React**, **SQL**, and **Azure**. Certified in **Microsoft Azure Fundamentals (AZ-900)**, with expertise in end-to-end development, API integration, and production support for high-volume platforms. Currently pursuing an **MSc in Software Engineering**, focusing on **machine learning** and AI-based projects, including a dissertation titled *"A Comparative Study of Convolutional Neural Networks for Dietary Analysis"*.

SKILLS

Experienced: .NET (C#), Ruby on Rails, React.js, HTML, CSS, JavaScript, Python, SQL

Familiar: Java, C/C++, Node.js

Tools: Git & GitHub, Azure (Event Hub, Service Bus, Functions), AWS, Heroku, MS Office

PROFESSIONAL EXPERIENCE

Engineer | Qualitest Infotech Pvt. Ltd. | Noida, India

Apr 2023 – Aug 2025

Client: Penn Foster Inc. | United States

Reason for leaving: Pursued a master's degree to strengthen expertise in software engineering and AI.

1. AiGrading Application

- Developed from scratch using **React**, **.NET (C#)**, and **SQL**, integrating with **D2L Brightspace (Learning Management System)** via **LTI 1.3** for seamless and secure single sign-on and grade passback.
- Integrated Learnosity's Feedback Aide API to deliver **AI-generated feedback** on student submissions, enabling iterative resubmission workflows that supported learner mastery, used by **over 75,000 learners**.
- Built a workflow routing final submissions to a proprietary grading platform for manual review and quality assurance.

2. Learning Management System (LMS) Connector for D2L Brightspace Replatforming

- **Developed independently** an **Azure** serverless **.NET** application to process events from **D2L Brightspace (LMS)** for a user base of **more than 500,000**, handling **1 to 2 million** events daily.
- Integrated **Event Hub** and **Service Bus** for data filtering and reliable message queuing, and synchronized grades with **Anthology Student Information System (SIS)**.

3. React Frontend Application with API Integration

- Built a **React** frontend for managing course, lesson, and program mappings, with backend database interaction via an HTTP trigger in LMS Connector, ensuring an intuitive user interface for data updates.

4. Proprietary Grading Platform

- Modified the **legacy** proprietary grading platform application to enhance usability and efficiency in grading student assignments for the replatformed LMS environment.

Graduate Engineer | Qualitest Infotech Pvt. Ltd. | Noida, India

Jan 2022 – Apr 2023

Client: Penn Foster Inc. | United States

1. PFX 2.0 / UNOW - Legacy LMS Maintenance

- Maintained a **Ruby on Rails** application deployed on **Heroku**, ensuring system **reliability** and **functionality**.

2. D2L Brightspace Print Application Optimization

- The D2L Print application serves as a widget within the **Desire2Learn Brightspace Learning Management System (LMS)**, enabling users to download course content in PDF format.

- **Working independently**, undertook a comprehensive **redesign** of the application to **enhance** its **efficiency** in generating PDFs.
- **Performance optimizations** were implemented to facilitate faster content downloads, significantly **improving** the application's **responsiveness**.

EDUCATION

MSc Software Engineering (Machine Learning) University of Limerick Limerick, Ireland 3.44 / 4 QCA (1 st Semester)	Sep 2025 – Aug 2026
BTech in Information Technology Noida Institute of Engineering & Technology Greater Noida, India 7.98 / 10 GPA (I-DIV WITH DISTINCTION)	Aug 2018 – Nov 2022

ACTIVITIES

Mentor Teen-Turn Programme University of Limerick Mentored teenage girls from underserved communities in STEM, supporting hands-on tech learning and career development
Participant Lero Hack 2026, Snapdragon: AI Unplugged Hackathon University of Limerick Competed in an AI-focused hackathon, designing and building an edge AI solution under time constraints

ACHIEVEMENTS

HighQ Award Winner – 2023 & 2024 Qualitest Infotech Pvt. Ltd. India Recognized for delivering high-quality solutions and exceeding client expectations two years in a row.
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CERTIFICATIONS

AZ-900 Microsoft Certified: Azure Fundamentals May 2025 (click)
.NET Development – Foundation Jan 2024 (click)
Introduction to Generative AI and Prompt Engineering June 2025 (click)